

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 1 of 40

Topic: NumPy Array Creation

Definition: NumPy Array Creation is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 1: Create a NumPy array from a Python list.

Input

```
[10, 20, 30]
```

Program

```
import numpy as np
arr = np.array([10, 20, 30])
print(arr)
```

Output

```
[10 20 30]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 2 of 40

Topic: Zero Array

Definition: Zero Array is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 2: Create an array of zeros.

Input

Shape (2, 3)

Program

```
import numpy as np
arr = np.zeros((2, 3), dtype=int)
print(arr)
```

Output

```
[[0 0 0]
 [0 0 0]]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 3 of 40

Topic: Ones Array

Definition: Ones Array is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 3: Create an array of ones.

Input

Shape (2, 2)

Program

```
import numpy as np
arr = np.ones((2, 2), dtype=int)
print(arr)
```

Output

```
[[1 1]
 [1 1]]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 4 of 40

Topic: NumPy arange()

Definition: NumPy arange() is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 4: Create a range array using arange().

Input

1 to 9

Program

```
import numpy as np
arr = np.arange(1, 10)
print(arr)
```

Output

[1 2 3 4 5 6 7 8 9]

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 5 of 40

Topic: NumPy linspace()

Definition: NumPy linspace() is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 5: Create evenly spaced values using linspace().

Input

0 to 1, 5 values

Program

```
import numpy as np
arr = np.linspace(0, 1, 5)
print(arr)
```

Output

```
[0.  0.25 0.5  0.75 1.  ]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 6 of 40

Topic: Array Shape

Definition: Array Shape is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 6: Find the shape of a 2D array.

Input

```
[[1, 2, 3], [4, 5, 6]]
```

Program

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
print(arr.shape)
```

Output

```
(2, 3)
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 7 of 40

Topic: Array Dimensions

Definition: Array Dimensions is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 7: Find the number of dimensions of an array.

Input

```
[[1,2],[3,4]]
```

Program

```
import numpy as np
arr = np.array([[1,2],[3,4]])
print(arr.ndim)
```

Output

```
2
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 8 of 40

Topic: Array Size

Definition: Array Size is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 8: Find the size of an array.

Input

```
[1, 2, 3, 4, 5]
```

Program

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5])
print(arr.size)
```

Output

```
5
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 9 of 40

Topic: Array Data Type

Definition: Array Data Type is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 9: Find the data type of an array.

Input

```
[1, 2, 3]
```

Program

```
import numpy as np
arr = np.array([1, 2, 3])
print(arr.dtype)
```

Output

```
int64 (platform dependent)
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 10 of 40

Topic: Array Shape

Definition: Array Shape is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 10: Reshape a 1D array into 2D.

Input

```
[1, 2, 3, 4, 5, 6]
```

Program

```
import numpy as np
arr = np.arange(1, 7).reshape(2, 3)
print(arr)
```

Output

```
[[1 2 3]
 [4 5 6]]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 11 of 40

Topic: Array Flattening

Definition: Array Flattening is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 11: Flatten a 2D array.

Input

```
[[1,2],[3,4]]
```

Program

```
import numpy as np
arr = np.array([[1,2],[3,4]])
print(arr.flatten())
```

Output

```
[1 2 3 4]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 12 of 40

Topic: Matrix Transpose

Definition: Matrix Transpose is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 12: Transpose a matrix.

Input

```
[[1, 2, 3], [4, 5, 6]]
```

Program

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
print(arr.T)
```

Output

```
[[1 4]
 [2 5]
 [3 6]]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 13 of 40

Topic: Array Sum

Definition: Array Sum is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 13: Find the sum of all elements.

Input

```
[1, 2, 3, 4]
```

Program

```
import numpy as np
arr = np.array([1, 2, 3, 4])
print(arr.sum())
```

Output

```
10
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 14 of 40

Topic: Array Mean

Definition: Array Mean is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 14: Find the mean of an array.

Input

```
[10, 20, 30]
```

Program

```
import numpy as np
arr = np.array([10, 20, 30])
print(arr.mean())
```

Output

```
20.0
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 15 of 40

Topic: Minimum and Maximum

Definition: Minimum and Maximum is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 15: Find minimum and maximum values.

Input

```
[7, 2, 9, 4]
```

Program

```
import numpy as np
arr = np.array([7, 2, 9, 4])
print(arr.min(), arr.max())
```

Output

```
2 9
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 16 of 40

Topic: Standard Deviation

Definition: Standard Deviation is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 16: Find the standard deviation.

Input

```
[1, 2, 3, 4]
```

Program

```
import numpy as np
arr = np.array([1, 2, 3, 4])
print(round(arr.std(), 4))
```

Output

```
1.1180
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 17 of 40

Topic: Element-wise Addition

Definition: Element-wise Addition is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 17: Add two arrays element-wise.

Input

```
[1, 2, 3] and [4, 5, 6]
```

Program

```
import numpy as np
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])
print(a + b)
```

Output

```
[5 7 9]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 18 of 40

Topic: Element-wise Multiplication

Definition: Element-wise Multiplication is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 18: Multiply two arrays element-wise.

Input

```
[2,3,4] and [5,6,7]
```

Program

```
import numpy as np
a = np.array([2,3,4])
b = np.array([5,6,7])
print(a * b)
```

Output

```
[10 18 28]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 19 of 40

Topic: Matrix Multiplication

Definition: Matrix Multiplication is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 19: Perform matrix multiplication.

Input

```
[[1,2],[3,4]] and [[5,6],[7,8]]
```

Program

```
import numpy as np
a = np.array([[1,2],[3,4]])
b = np.array([[5,6],[7,8]])
print(a @ b)
```

Output

```
[[19 22]
 [43 50]]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 20 of 40

Topic: Boolean Indexing

Definition: Boolean Indexing is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 20: Select elements greater than 5.

Input

```
[2, 6, 3, 8, 1]
```

Program

```
import numpy as np
arr = np.array([2, 6, 3, 8, 1])
print(arr[arr > 5])
```

Output

```
[6 8]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 21 of 40

Topic: Boolean Indexing

Definition: Boolean Indexing is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 21: Replace values greater than 5 with 0.

Input

```
[2, 6, 3, 8, 1]
```

Program

```
import numpy as np
arr = np.array([2, 6, 3, 8, 1])
arr[arr > 5] = 0
print(arr)
```

Output

```
[2 0 3 0 1]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 22 of 40

Topic: Non-zero Indices

Definition: Non-zero Indices is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 22: Find indices of non-zero elements.

Input

```
[0, 4, 0, 7]
```

Program

```
import numpy as np
arr = np.array([0, 4, 0, 7])
print(np.nonzero(arr)[0])
```

Output

```
[1 3]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 23 of 40

Topic: Array Sorting

Definition: Array Sorting is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 23: Sort an array.

Input

```
[5, 2, 9, 1]
```

Program

```
import numpy as np
arr = np.array([5, 2, 9, 1])
print(np.sort(arr))
```

Output

```
[1 2 5 9]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 24 of 40

Topic: Unique Values

Definition: Unique Values is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 24: Get unique values.

Input

```
[1, 2, 2, 3, 3, 3]
```

Program

```
import numpy as np
arr = np.array([1, 2, 2, 3, 3, 3])
print(np.unique(arr))
```

Output

```
[1 2 3]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 25 of 40

Topic: Array Concatenation

Definition: Array Concatenation is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 25: Concatenate two arrays.

Input

```
[1,2] and [3,4]
```

Program

```
import numpy as np
a = np.array([1,2])
b = np.array([3,4])
print(np.concatenate([a,b]))
```

Output

```
[1 2 3 4]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 26 of 40

Topic: Array Splitting

Definition: Array Splitting is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 26: Split an array into two parts.

Input

```
[1, 2, 3, 4]
```

Program

```
import numpy as np
arr = np.array([1, 2, 3, 4])
print(np.split(arr, 2))
```

Output

```
[array([1, 2]), array([3, 4])]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 27 of 40

Topic: Random Integer Generation

Definition: Random Integer Generation is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 27: Generate random integers.

Input

Seed 1, two values from 1 to 10

Program

```
import numpy as np
np.random.seed(1)
print(np.random.randint(1, 11, 2))
```

Output

[6 9]

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 28 of 40

Topic: Identity Matrix

Definition: Identity Matrix is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 28: Create an identity matrix.

Input

3 x 3

Program

```
import numpy as np
print(np.eye(3, dtype=int))
```

Output

```
[[1 0 0]
 [0 1 0]
 [0 0 1]]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 29 of 40

Topic: Row-wise Sum

Definition: Row-wise Sum is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 29: Find row-wise sums.

Input

```
[[1,2],[3,4]]
```

Program

```
import numpy as np
arr = np.array([[1,2],[3,4]])
print(arr.sum(axis=1))
```

Output

```
[3 7]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 30 of 40

Topic: Column-wise Sum

Definition: Column-wise Sum is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 30: Find column-wise sums.

Input

```
[[1,2],[3,4]]
```

Program

```
import numpy as np
arr = np.array([[1,2],[3,4]])
print(arr.sum(axis=0))
```

Output

```
[4 6]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 31 of 40

Topic: NumPy Broadcasting

Definition: NumPy Broadcasting is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 31: Broadcast a scalar across an array.

Input

```
[1,2,3] + 10
```

Program

```
import numpy as np
arr = np.array([1,2,3])
print(arr + 10)
```

Output

```
[11 12 13]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 32 of 40

Topic: Integer to Float Conversion

Definition: Integer to Float Conversion is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 32: Convert an integer array to float.

Input

```
[1, 2, 3]
```

Program

```
import numpy as np
arr = np.array([1, 2, 3])
print(arr.astype(float))
```

Output

```
[1.  2.  3.]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 33 of 40

Topic: Cumulative Sum

Definition: Cumulative Sum is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 33: Calculate cumulative sum.

Input

```
[1, 2, 3, 4]
```

Program

```
import numpy as np
arr = np.array([1, 2, 3, 4])
print(arr.cumsum())
```

Output

```
[1 3 6 10]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 34 of 40

Topic: Cumulative Product

Definition: Cumulative Product is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 34: Calculate cumulative product.

Input

```
[1, 2, 3, 4]
```

Program

```
import numpy as np
arr = np.array([1, 2, 3, 4])
print(arr.cumprod())
```

Output

```
[1 2 6 24]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 35 of 40

Topic: Median

Definition: Median is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 35: Find the median.

Input

```
[1, 7, 3, 9, 5]
```

Program

```
import numpy as np
arr = np.array([1, 7, 3, 9, 5])
print(np.median(arr))
```

Output

```
5.0
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 36 of 40

Topic: Variance

Definition: Variance is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 36: Find variance.

Input

```
[1, 2, 3, 4]
```

Program

```
import numpy as np
arr = np.array([1, 2, 3, 4])
print(arr.var())
```

Output

```
1.25
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 37 of 40

Topic: 2D Array Indexing

Definition: 2D Array Indexing is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 37: Index a 2D array.

Input

```
[[10,20],[30,40]], row 2 col 1
```

Program

```
import numpy as np
arr = np.array([[10,20],[30,40]])
print(arr[1,0])
```

Output

```
30
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 38 of 40

Topic: Array Slicing

Definition: Array Slicing is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 38: Slice rows from a 2D array.

Input

```
[[1,2],[3,4],[5,6]]
```

Program

```
import numpy as np
arr = np.array([[1,2],[3,4],[5,6]])
print(arr[1:])
```

Output

```
[[3 4]
 [5 6]]
```

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 39 of 40

Topic: Dot Product

Definition: Dot Product is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 39: Find the dot product of two arrays.

Input

[1, 2, 3] and [4, 5, 6]

Program

```
import numpy as np
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])
print(np.dot(a, b))
```

Output

32

TOP 40 NUMPY INTERVIEW QUESTIONS

Question 40 of 40

Topic: NaN Replacement

Definition: NaN Replacement is a NumPy technique used to work efficiently with numerical arrays. It helps perform the required array operation with concise vectorized code.

Question 40: Replace NaN with zero.

Input

```
[1, NaN, 3]
```

Program

```
import numpy as np
arr = np.array([1, np.nan, 3])
arr = np.nan_to_num(arr, nan=0)
print(arr)
```

Output

```
[1.  0.  3.]
```